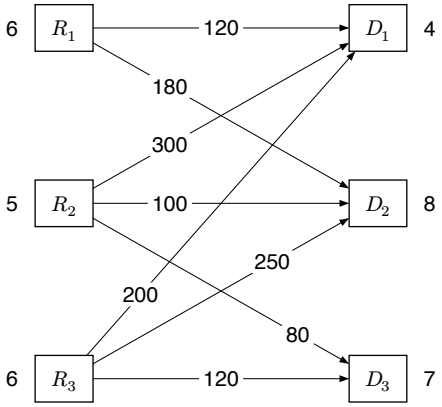
Carmen Carrera
Divya Bhojane
Vladimir Borel

Problem 1



Sets

- $I = \{1, 2, 3\}$: set of **refineries** (source nodes)
- $J = \{1, 2, 3\}$: set of **distribution areas** (sink nodes)
- $J^S = \{2, 3\} \subset J$: areas where a **shortage is allowed** (penalized by p)
- $A = I \times J \setminus \{(1, 3)\}$: set of exisiting **pipelines** (arcs)

Indices

- $i \in I$: **refinery**
- $j \in J$: **distribution area**

Parameters

- S_i: **supply** capacity of refinery i (Mgal)

$$S = \begin{bmatrix} 6 \\ 5 \\ 6 \end{bmatrix}$$

- D_j: **demand** of area j (Mgal)

$$D = \begin{bmatrix} 4 \\ 8 \\ 7 \end{bmatrix}$$

- m_{ij}: **distance** from refinery i to area j (miles)

$$M = \begin{bmatrix} 120 & 180 & - \\ 300 & 100 & 80 \\ 200 & 250 & 120 \end{bmatrix}$$

- $c_{ij} = 10 m_{ij}$: unit transportation **cost** (k$/Mgal) ($1 / 100 gal / mile)
- $p = 5000$: shortage **penalty** (k$/Mgal) ($5 per gal)

Decision variables

- $x_{ij} \geq 0 \quad \forall (i, j) \in A$: **quantity** of oil shipped from refinery i to area j (Mgal)
- $s_j \geq 0 \quad \forall j \in J^S$: **shortage** at area j (Mgal)

Model

$$
\begin{aligned}
\min \quad & \sum_{i \in I} \sum_{j \in J} c_{ij} x_{ij} + p \sum_{j \in J^S} s_j \\
\text{s.t.} \quad & \sum_{j \in J} x_{ij} \le S_i && \forall i \in I && \text{capacity} \\
& \sum_{i \in I} x_{ij} + s_j [j \in J^S] = D_j && \forall j \in J && \text{demand} \\
& x_{ij} \ge 0 && \forall (i,j) \in A \\
& s_j \ge 0 && \forall j \in J^S
\end{aligned}
$$

$$
\begin{aligned}
\min \quad & c_{11}x_{11} + c_{12}x_{12} + \\
& c_{21}x_{21} + c_{22}x_{22} + c_{23}x_{23} + \\
& c_{31}x_{31} + c_{32}x_{32} + c_{33}x_{33} \\
& + p(s_2 + s_3) \\
\text{s.t.} \quad & \left.\begin{aligned} x_{11} + x_{12} &\le S_1 \\ x_{21} + x_{22} + x_{23} &\le S_2 \\ x_{31} + x_{32} + x_{33} &\le S_3 \end{aligned}\right\} \text{ capacity} \\
& \left.\begin{aligned} x_{11} + x_{21} + x_{31} &= D_1 \\ x_{12} + x_{22} + x_{32} + s_2 &= D_2 \\ x_{23} + x_{33} + s_3 &= D_3 \end{aligned}\right\} \text{ demand} \\
& x_{11}, x_{12}, x_{21}, x_{22}, x_{23}, x_{31}, x_{32}, x_{33} \ge 0 \\
& s_2, s_3 \ge 0
\end{aligned}
$$

Code

See `problem_1.py`.

Solution

Objective	30400

Variable	Value	Reduced Cost	Objective Coef	Allowable Increase	Allowable Decrease
x[1,1]	4	0	1200	1200	∞
x[1,2]	2	0	1800	3200	1200
x[2,1]	0	2600	3000	∞	2600
x[2,2]	4	0	1000	1100	200
x[2,3]	1	0	800	200	1100
x[3,1]	0	1200	2000	∞	1200
x[3,2]	0	1100	2500	∞	1100
x[3,3]	6	0	1200	1100	∞
s[2]	2	0	5000	200	3200
s[3]	0	200	5000	∞	200

Table 1: Variables

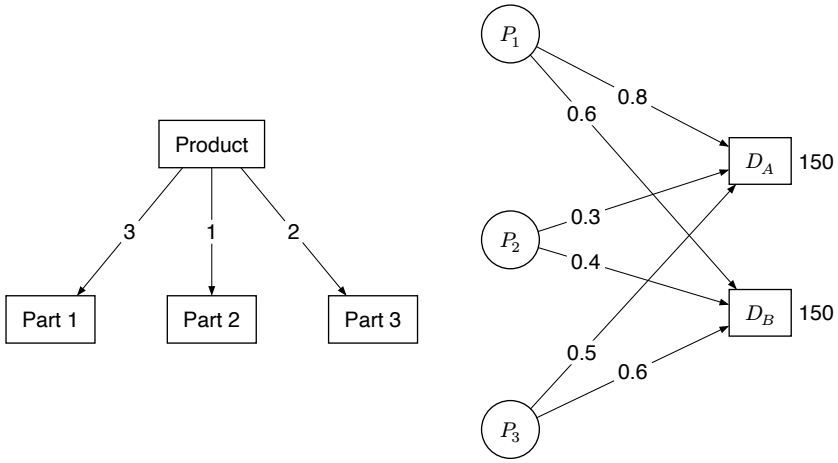
Name	Value	Shadow Price	RHS	Slack	Allowable Increase	Allowable Decrease
capacity[1]	6	–3200	6	0	2	2
capacity[2]	5	–4000	5	0	2	4
capacity[3]	6	–3600	6	0	1	4
demand[1]	4	4400	4	0	2	2
demand[2]	8	5000	8	0	∞	2
demand[3]	7	4800	7	0	4	1

Table 2: Constraints

Edge	Cost
1,1	1200
1,2	1800
2,1	3000
2,2	1000
2,3	800
3,1	2000
3,2	2500
3,3	1200

Table 3: Cost

Problem 2



Sets

- $I = \{1, 2, 3\}$: set of **parts**
- $J = \{A, B\}$: set of **departments**
- $A = I \times J$

Indices

- $i \in I$: **parts**
- $j \in J$: **departments**

Parameters

- t_{ij}: time it takes to produce **part** i in **departement** j

$$T = \begin{bmatrix} 0.8 & 0.6 \\ 0.3 & 0.4 \\ 0.5 & 0.6 \end{bmatrix}$$

- S_j: available production time in **departement** j

$$S = \begin{bmatrix} 150 \\ 150 \end{bmatrix}$$

- r_i: amount of **part** i it takes to create a product

$$R = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$$

Decision variables

- y: number of **products** produced
- x_{ij}: how many **parts** i are produced in **department** j

Model

$$
\begin{aligned}
\max \quad & y \\
\text{s.t.} \quad & \sum_{j \in J} x_{ij} \geq r_i y && \forall i \in I \quad \text{requirements} \\
& \sum_{i \in I} t_{ij} x_{ij} \leq S_j && \forall j \in J \quad \text{capacity} \\
& x_{ij} \geq 0 && \forall i \in I, j \in J \\
& y \geq 0
\end{aligned}
$$

$$
\begin{aligned}
\max \quad & y \\
\text{s.t.} \quad & \left.\begin{aligned} x_{1A} + x_{1B} &\geq 3y \\ x_{2A} + x_{2B} &\geq y \\ x_{3A} + x_{3B} &\geq 2y \end{aligned}\right\} \quad \text{requirements} \\
& \left.\begin{aligned} 0.8x_{1A} + 0.3x_{2A} + 0.5x_{3A} &\leq 150 \\ 0.6x_{1B} + 0.4x_{2B} + 0.6x_{3B} &\leq 150 \end{aligned}\right\} \quad \text{capacity} \\
& x_{1A}, x_{1B}, x_{2A}, x_{2B}, x_{3A}, x_{3B} \geq 0 \\
& y \geq 0
\end{aligned}
$$

Code

See `problem_2.py`.

Solution

Objective	94.5946

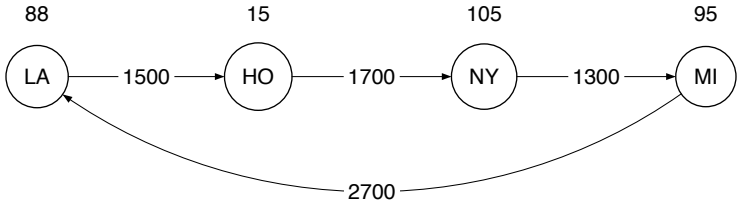
Variable	Value	Reduced Cost	Objective Coef	Allowable Increase	Allowable Decrease
x[1,1]	33.7838	0	0	0.1071	0.3333
x[1,2]	250	0	0	∞	0.0811
x[2,1]	94.5946	0	0	0.0882	0.0593
x[2,2]	0	–0.0631	0	0.0631	∞
x[3,1]	189.1892	0	0	0.1852	0.0698
x[3,2]	0	–0.0811	0	0.0811	∞
y	94.5946	0	1	∞	1

Table 4: Variables

Name	Value	Shadow Price	RHS	Slack	Allowable Increase	Allowable Decrease
requirements[1]	0	–0.2162	0	0	437.5	96.1538
requirements[2]	0	–0.0811	0	0	138.8889	102.9412
requirements[3]	0	–0.1351	0	0	83.3333	259.2593
capacity[1]	150	0.2703	150	0	∞	41.6667
capacity[2]	150	0.3604	150	0	57.6923	150

Table 5: Constraints

Problem 3



Sets

- $I = \{1, 2, 3, 4\}$

Indices

- $i \in I$: **airports**

Indices cyclic ($i = 5 \equiv 1$, $i = 0 \equiv 4$)

Parameters

- m_i: distance between city i and city $i + 1$ (miles)

$$M = \begin{bmatrix} 1500 \\ 1700 \\ 1300 \\ 2700 \end{bmatrix}$$

- p_i: **price** per gallon (¢/gal)

$$P = \begin{bmatrix} 88 \\ 15 \\ 105 \\ 95 \end{bmatrix}$$

- $k = 12000$: plane fuel **capacity** (gallons)
- $r = 600$: fuel **reserve** (gallons)

Decision Variables

- $x_i \geq 0$: how much fuel to put into the plane at airport i (gallons)
- $s_i \geq 0$: fuel level at start of leg i, after refuelling at airport i (gallons)
- $e_i \geq 0$: fuel level at end of leg i, on arrival at airport $i+1$ before refuelling (gallons)

Fuel-burn relation

$$s_i - e_i = m_i \left(1 + \frac{(s_i + e_i)/2}{2000}\right)$$

Model

$$\begin{aligned}
\min \quad & \sum_{i \in I} p_i x_i & & \\
\text{s.t.} \quad & s_i = e_{i-1} + x_i & \forall i \in I \quad & \text{balance} \\
& s_i - e_i = m_i \left(1 + \frac{s_i + e_i}{4000}\right) & \forall i \in I \quad & \text{consumption} \\
& s_i \leq k & \forall i \in I \quad & \text{capacity} \\
& e_i \geq r & \forall i \in I \quad & \text{reserve} \\
& x_i \geq 0 & \forall i \in I & \\
& s_i \geq 0 & \forall i \in I & \\
& e_i \geq 0 & \forall i \in I &
\end{aligned}$$

$$\begin{aligned}
\min \quad & p_1 x_1 + p_2 x_2 + p_3 x_3 + p_4 x_4 \\
\text{s.t.} \quad & \\
& \left.\begin{aligned} s_2 &= e_1 + x_2 \\ s_3 &= e_2 + x_3 \\ s_4 &= e_3 + x_4 \\ s_1 &= e_4 + x_1 \end{aligned}\right\} \text{ balance} \\
& \left.\begin{aligned} s_1 - e_1 &= m_1\left(1 + \tfrac{s_1+e_1}{4000}\right) \\ s_2 - e_2 &= m_2\left(1 + \tfrac{s_2+e_2}{4000}\right) \\ s_3 - e_3 &= m_3\left(1 + \tfrac{s_3+e_3}{4000}\right) \\ s_4 - e_4 &= m_4\left(1 + \tfrac{s_4+e_4}{4000}\right) \end{aligned}\right\} \text{ consumption} \\
& \left.\begin{aligned} s_1 &\leq k \\ s_2 &\leq k \\ s_3 &\leq k \\ s_4 &\leq k \end{aligned}\right\} \text{ capacity} \\
& \left.\begin{aligned} e_1 &\geq r \\ e_2 &\geq r \\ e_3 &\geq r \\ e_4 &\geq r \end{aligned}\right\} \text{ reserve} \\
& x_1, x_2, x_3, x_4 \geq 0 \\
& s_1, s_2, s_3, s_4 \geq 0 \\
& e_1, e_2, e_3, e_4 \geq 0
\end{aligned}$$

Code

See `problem_3.py`.

Solution

Objective	1445163.7736

Variable	Value	Reduced Cost	Objective Coef	Allowable Increase	Allowable Decrease
x[1]	3120	0	88	∞	81.1818
x[2]	11400	0	15	4.5283	∞
x[3]	0	56.6038	105	∞	56.6038
x[4]	10522.145	0	95	111.1111	22.029
s[1]	3720	0	0	∞	81.1818
s[2]	12000	0	0	4.5283	∞
s[3]	3649.1228	0	0	11.2223	56.6038
s[4]	11400	0	0	∞	95
e[1]	600	0	0	∞	178.6
e[2]	3649.1228	0	0	11.2223	∞
e[3]	877.855	0	0	22.029	111.1111
e[4]	600	0	0	∞	489.6154

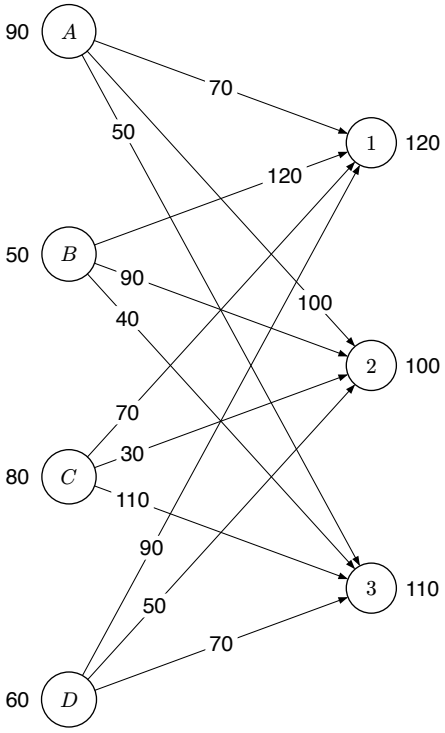
Table 6: Variables

Name	Value	Shadow Price	RHS	Slack	Allowable Increase	Allowable Decrease
balance[1]	600	–88	600	0	3120	∞
balance[2]	0	–15	0	0	11400	∞
balance[3]	0	–48.3962	0	0	8350.8772	545.4191
balance[4]	0	–95	0	0	10522.145	∞
consumption[1]	1500	140.8	1500	0	5175	1950
consumption[2]	1700	33.9623	1700	0	777.2222	11900
consumption[3]	1300	71.6981	1300	0	368.1579	13941.8421
consumption[4]	2700	292.3077	2700	0	195	3419.6971
capacity[1]	3720	0	12000	8280	∞	8280
capacity[2]	12000	–4.5283	12000	0	20695.6522	1351.6908
capacity[3]	3649.1228	0	12000	8350.8772	∞	8350.8772
capacity[4]	11400	0	12000	600	∞	600
reserve[1]	600	178.6	600	0	3763.6364	600
reserve[2]	3649.1228	0	600	3049.1228	3049.1228	∞
reserve[3]	877.855	0	600	277.855	277.855	∞
reserve[4]	600	489.6154	600	0	116.4179	600

Table 7: Constraints

- The model takes advantage of the low fuel price in Houston ($0.15/gal) by filling the tank completely (12,000 gallons). Even though the plane burns more fuel on the Houston-NYC leg because it is heavier, it is still cheaper than buying fuel elsewhere.
- No fuel is bought in NYC, the most expensive airport, since the fuel from Houston is enough to reach Miami.
- In LA and Miami the plane only buys the fuel needed to arrive at the next airport with the minimum 600 gallons, because carrying extra fuel increases consumption.
- The total fuel cost per round trip is $14,451.64.

Problem 4



Sets

- $I = \{1, 2, 3, 4\}$: set of **supply** nodes
- $J = \{1, 2, 3\}$: set of **demand** nodes

Indices

- $i \in I$: **supply** nodes
- $j \in J$: **demand** nodes

Parameters

- k_i: **capacity** at supply node i

$$K = \begin{bmatrix} 90 \\ 50 \\ 80 \\ 60 \end{bmatrix}$$

- d_j: **demand** at demand node j

$$D = \begin{bmatrix} 120 \\ 100 \\ 110 \end{bmatrix}$$

- c_{ij}: unit **cost** of transporting a product from supply node i to demand node j

$$C = \begin{bmatrix} 70 & 100 & 50 \\ 120 & 90 & 40 \\ 70 & 30 & 110 \\ 90 & 50 & 70 \end{bmatrix}$$

Decision Variables

- x_{ij}: number of units sent from **supply node** i to **demand node** j

Model

$$
\begin{aligned}
\min \quad & \sum_{i \in I} \sum_{j \in J} c_{ij} x_{ij} \\
\text{s.t.} \quad & \sum_{j \in J} x_{ij} = k_i && \forall i \in I \quad \text{supply} \\
& \sum_{i \in I} x_{ij} \leq d_j && \forall j \in J \quad \text{demand} \\
& x_{ij} \geq 0 \forall i \in I, j \in J
\end{aligned}
$$

$$
\begin{aligned}
\min \quad & c_{11}x_{11} + c_{12}x_{12} + c_{13}x_{13} + \\
& c_{21}x_{21} + c_{22}x_{22} + c_{23}x_{23} + \\
& c_{31}x_{31} + c_{32}x_{32} + c_{33}x_{33} + \\
& c_{41}x_{41} + c_{42}x_{42} + c_{43}x_{43} \\
\text{s.t.} \quad & \\
& \left.\begin{aligned}
x_{11} + x_{12} + x_{13} &= k_1 \\
x_{21} + x_{22} + x_{23} &= k_2 \\
x_{31} + x_{32} + x_{33} &= k_3 \\
x_{41} + x_{42} + x_{43} &= k_4
\end{aligned}\right\} \text{supply} \\
& \left.\begin{aligned}
x_{11} + x_{21} + x_{31} + x_{41} &\leq d_1 \\
x_{12} + x_{22} + x_{32} + x_{42} &\leq d_2 \\
x_{13} + x_{23} + x_{33} + x_{43} &\leq d_3
\end{aligned}\right\} \text{demand} \\
& x_{11}, x_{12}, x_{13}, \\
& x_{21}, x_{22}, x_{23}, \\
& x_{31}, x_{32}, x_{33}, \\
& x_{41}, x_{42}, x_{43} \geq 0
\end{aligned}
$$

Code

See `problem_4.py`.

Solution

Objective	14100

Variable	Value	Reduced Cost	Objective Coef	Allowable Increase	Allowable Decrease
x[0,0]	70	0	70	0	20
x[0,1]	0	70	100	∞	70
x[0,2]	20	0	50	20	0
x[1,0]	0	60	120	∞	60
x[1,1]	0	70	90	∞	70
x[1,2]	50	0	40	60	∞
x[2,0]	0	0	70	∞	0
x[2,1]	80	0	30	0	∞
x[2,2]	0	60	110	∞	60
x[3,0]	0	0	90	∞	0
x[3,1]	20	0	50	40	0
x[3,2]	40	0	70	0	40

Table 8: Variables

Name	Sense	Value	Shadow Price	RHS	Slack	Allowable Increase	Allowable Decrease
supply[0]	==	90	70	90	0	50	70
supply[1]	==	50	60	50	0	20	50
supply[2]	==	80	70	80	0	20	40
supply[3]	==	60	90	60	0	20	40
demand[0]	<=	70	0	120	50	∞	50
demand[1]	<=	100	−40	100	0	40	20
demand[2]	<=	110	−20	110	0	70	20

Table 9: Constraints

Edge	Cost
0,0	70
0,1	100
0,2	50
1,0	120
1,1	90
1,2	40
2,0	70
2,1	30
2,2	110
3,0	90
3,1	50
3,2	70

Table 10: Cost

2.

Check the allowable range. The change is –10 and the allowable decrease is 20. Since 10 ≤ 20, the current basis stays optimal. Use the shadow price to get the cost change. The shadow price of –20 means each extra ton of Durham demand lowers total cost by \$20. That happens because each ton Durham takes can be shipped there cheaply instead of somewhere more expensive. Going the other way, removing 10 tons raises cost:

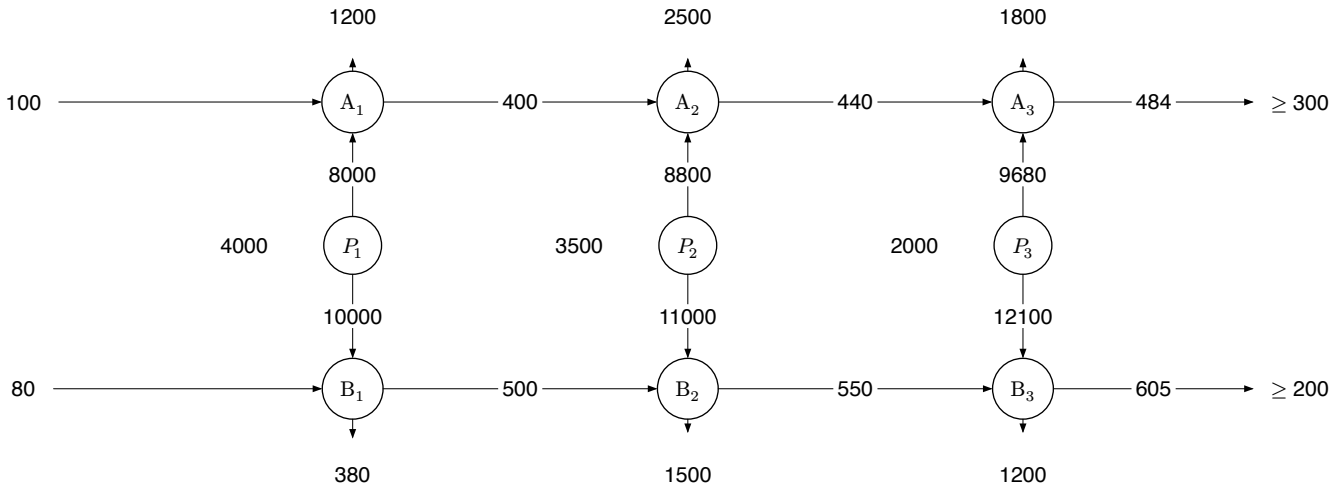
$$\Delta \text{ Cost} = (-20)x(-10) = 200$$

The routes and basis don't change, but the shipment amounts and the cost do. New total cost (\$14,300) and the new shipments (Charlotte → Richmond 80, Charlotte → Durham 10).

3.

Lexington's shadow price is \$70 and holds for up to 20 extra tons, so each added ton serves one more ton of Richmond's unmet demand at \$70 of extra shipping cost. Raleigh is cheaper at \$60 per ton (also up to 20 tons), and Charlotte matches Lexington at \$70 but holds for 50 tons, enough to cover Richmond's whole shortfall. So the numbers argue against the board's choice: add capacity at Raleigh first, then Charlotte, and avoid Danville at \$90.

Problem 5



Quarter	Production cost A21	Production cost B32	Holding cost A21	Holding cost B32	Demand A21	Demand B32	Capacity
1	8000	10000	400	500	1200	380	4000
2	8800	11000	440	550	2500	1500	3500
3	9680	12100	484	605	1800	1200	2000

Sets

- $I = \{1, 2\}$: set of **boat models**
- $J = \{1, 2, 3\}$: set of **quarters**

Indices

- $i \in I$
- $j \in J$

Parameters

y_i^0: **starting inventory** of model i

$$Y_0 = \begin{bmatrix} 100 \\ 80 \end{bmatrix}$$

- h_{ij}: **holding cost** of model i produced in **quarter** j

$$H = \begin{bmatrix} 400 & 500 \\ 440 & 550 \\ 484 & 605 \end{bmatrix}$$

- c_{ij}: production **cost** of model i produced in **quarter** j

$$C = \begin{bmatrix} 8000 & 10000 \\ 8800 & 11000 \\ 9680 & 12100 \end{bmatrix}$$

- d_{ij}: **demand** for model i in **quarter** j

$$D = \begin{bmatrix} 1200 & 380 \\ 2500 & 1500 \\ 1800 & 1200 \end{bmatrix}$$

- k_j: production **capacity** in quarter j

$$K = \begin{bmatrix} 4000 \\ 3500 \\ 2000 \end{bmatrix}$$

- e_i: end inventory for model i

$$E = \begin{bmatrix} 300 \\ 200 \end{bmatrix}$$

Decision Variables

- x_{ij}: number of units of **boat model** i to produce in **quarter** j
- y_{ij}: **inventory** of boats i at the end of **quarter** j

Model

$$
\begin{aligned}
\min \quad & \sum_{i \in I} \sum_{j \in J} (c_{ij} x_{ij} + h_{ij} y_{ij}) \\
\text{s.t.} \quad & y_{ij} = y_{i,j-1} + x_{ij} - d_{ij} && \forall i \in I, j \in J && \text{balance} \\
& \sum_{i \in I} x_{ij} \leq k_j && \forall j \in J && \text{capacity} \\
& y_{i3} \geq e_i && \forall i \in I && \text{end inventory} \\
& y_{i0} = y_i^0 && \forall i \in I && \text{initial inventory} \\
& x_{ij}, y_{ij} \geq 0 && \forall i \in I, j \in J
\end{aligned}
$$

$$
\begin{aligned}
\min \quad & x_{11}c_{11} + x_{12}c_{12} + x_{13}c_{13} + \\
& x_{21}c_{21} + x_{22}c_{22} + x_{23}c_{23} + \\
& y_{11}h_{11} + y_{12}h_{12} + y_{13}h_{13} + \\
& y_{21}h_{21} + y_{22}h_{22} + y_{23}h_{23} \\
\text{s.t.} \quad & \left.\begin{aligned}
& y_{11} = y_1^0 + x_{11} - d_{11} \\
& y_{12} = y_{11} + x_{12} - d_{12} \\
& y_{13} = y_{12} + x_{13} - d_{13} \\
& y_{21} = y_2^0 + x_{21} - d_{21} \\
& y_{22} = y_{21} + x_{22} - d_{22} \\
& y_{23} = y_{22} + x_{23} - d_{23}
\end{aligned}\right\} \text{balance} \\
& \left.\begin{aligned}
& x_{11} + x_{21} \leq k_1 \\
& x_{12} + x_{22} \leq k_2 \\
& x_{13} + x_{23} \leq k_3
\end{aligned}\right\} \text{capacity} \\
& \left.\begin{aligned}
& y_{12} + x_{13} - d_{13} \geq e_1 \\
& y_{22} + x_{23} - d_{23} \geq e_2
\end{aligned}\right\} \text{end inventory} \\
& x_{11}, x_{12}, x_{13}, \\
& x_{21}, x_{22}, x_{23} \geq 0 \\
& y_{11}, y_{12}, y_{13}, \\
& y_{21}, y_{22}, y_{23} \geq 0
\end{aligned}
$$

Code

See `problem_5.py`.

Solution

Objective	85456200

Variable	Value	Reduced Cost	Objective Coef	Allowable Increase	Allowable Decrease
x[0,0]	1100	0	8000	∞	100
x[0,1]	2900	0	10000	100	∞
x[1,0]	3200	0	8800	100	110
x[1,1]	300	0	11000	110	100
x[2,0]	1400	0	9680	110	440
x[2,1]	0	110	12100	∞	110
y[0,0]	0	100	400	∞	100
y[0,1]	2600	0	500	100	∞
y[1,0]	700	0	440	440	110
y[1,1]	1400	0	550	110	12595
y[2,0]	300	0	484	∞	10164
y[2,1]	200	0	605	∞	12595

Table 11: Variables

Name	Sense	Value	Shadow Price	RHS	Slack	Allowable Increase	Allowable Decrease
balance_q1[0]	==	–1100	–8940	–1100	0	300	600
balance_q1[1]	==	–300	–10940	–300	0	300	600
balance[0,0]	==	–2500	–9240	–2500	0	1400	600
balance[0,1]	==	–1500	–11440	–1500	0	300	600
balance[1,0]	==	–1800	–9680	–1800	0	1400	600
balance[1,1]	==	–1200	–11990	–1200	0	300	600
capacity[0]	<=	4000	–940	4000	0	300	600
capacity[1]	<=	3500	–440	3500	0	1400	600
capacity[2]	<=	1400	0	2000	600	∞	600
end_inventory[0]	>=	300	10164	300	0	600	300
end_inventory[1]	>=	200	12595	200	0	600	200

Table 12: Constraints

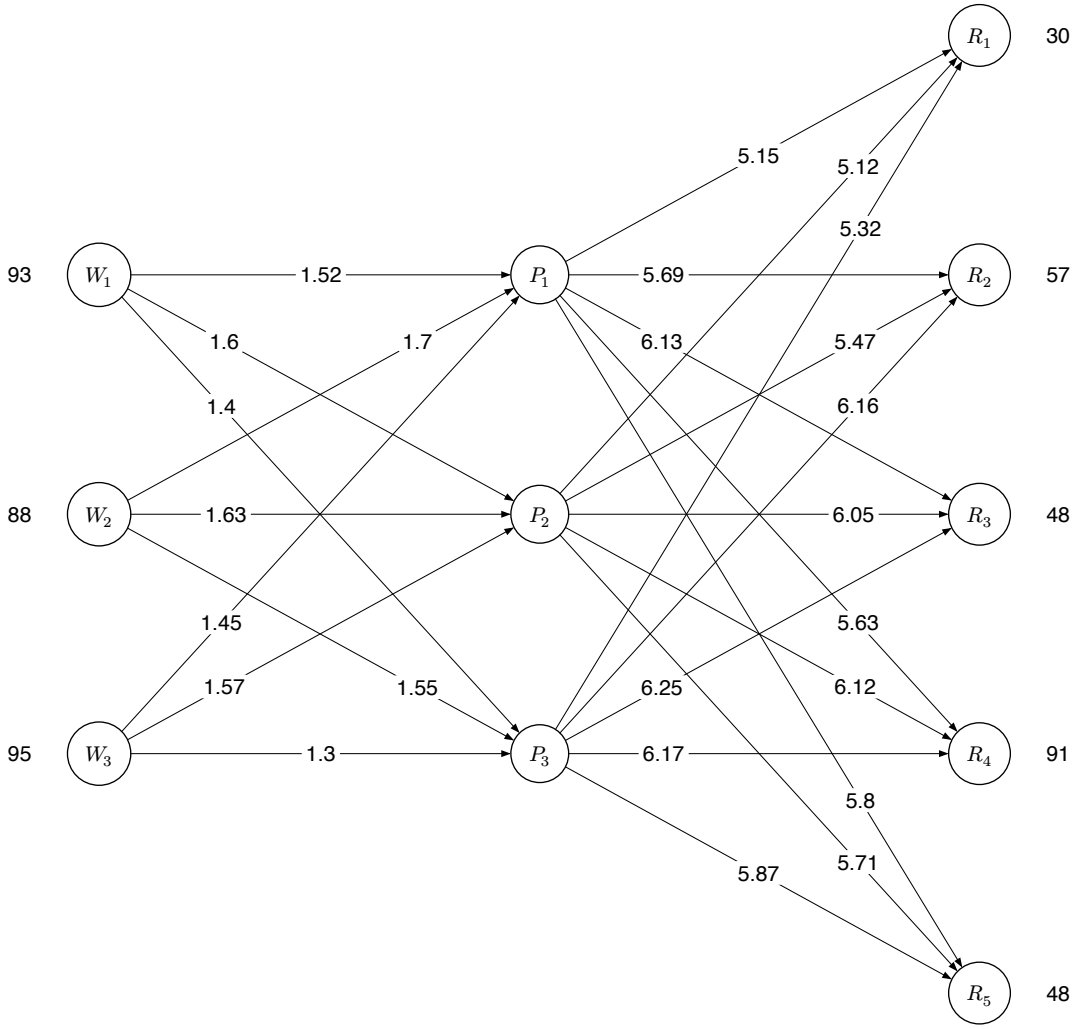
Quarter	A21	B32
Q1	8000	10000
Q2	8800	11000
Q3	9680	12100

Table 13: Production cost

Quarter	Produce A21	Produce B32	Inventory A21	Inventory B32
Q1	1100	2900	0	2600
Q2	3200	300	700	1400
Q3	1400	0	300	200

Table 14: Production plan

Problem 6



Sets

- $I = \{1, 2, 3\}$
- $J = \{1, 2, 3\}$
- $K = \{1, 2, 3, 4, 5\}$

Indices

- $i \in I$: set of **wells**
- $j \in J$: set of **pumps**
- $k \in K$: set of **refineries**

Parameters

- c_{ij}: cost of transport from well i to pump j

$$C^{IJ} = \begin{bmatrix} 1.52 & 1.60 & 1.40 \\ 1.70 & 1.63 & 1.55 \\ 1.45 & 1.57 & 1.30 \end{bmatrix}$$

- r_{jk}: cost of transport from pump j to refineries k

$$C^{JK} = \begin{bmatrix} 5.15 & 5.69 & 6.13 & 5.63 & 5.80 \\ 5.12 & 5.47 & 6.05 & 6.12 & 5.71 \\ 5.32 & 6.16 & 6.25 & 6.17 & 5.87 \end{bmatrix}$$

- S_i: supply at well i

$$S = \begin{bmatrix} 93 \\ 88 \\ 95 \end{bmatrix}$$

$$D = \begin{bmatrix} 30 \\ 57 \\ 48 \\ 91 \\ 48 \end{bmatrix}$$

Decision Variables

- x_{ij}: quantity flowing from well i to pump j
- y_{jk}: quantity flowing from pump j to refinery k

Model

$$\begin{aligned}
\min \quad & \sum_{i \in I} \sum_{j \in J} c_{ij} x_{ij} + \sum_{j \in J} \sum_{k \in K} r_{jk} y_{jk} \\
\text{s.t.} \quad & \sum_{j \in J} x_{ij} \leq S_i && \forall i \in I && \text{supply} \\
& \sum_{j \in J} y_{jk} = D_k && \forall k \in K && \text{demand} \\
& \sum_{i \in I} x_{ij} = \sum_{k \in K} y_{jk} && \forall j \in J && \text{conservation} \\
& x_{ij} \geq 0 \\
& y_{jk} \geq 0
\end{aligned}$$

$$\begin{aligned}
\min \quad & c_{11}x_{11} + c_{12}x_{12} + c_{13}x_{13} + \\
& c_{21}x_{21} + c_{22}x_{22} + c_{23}x_{23} + \\
& c_{31}x_{31} + c_{32}x_{32} + c_{33}x_{33} + \\
& r_{11}y_{11} + r_{12}y_{12} + r_{13}y_{13} + r_{14}y_{14} + r_{15}y_{15} + \\
& r_{21}y_{21} + r_{22}y_{22} + r_{23}y_{23} + r_{24}y_{24} + r_{25}y_{25} + \\
& r_{31}y_{31} + r_{32}y_{32} + r_{33}y_{33} + r_{34}y_{34} + r_{35}y_{35} \\
\text{s.t.} \quad & \left.\begin{aligned} x_{11} + x_{12} + x_{13} &\leq S_1 \\ x_{21} + x_{22} + x_{23} &\leq S_2 \\ x_{31} + x_{32} + x_{33} &\leq S_3 \end{aligned}\right\} \text{ supply} \\
& \left.\begin{aligned} y_{11} + y_{21} + y_{31} &= D_1 \\ y_{12} + y_{22} + y_{32} &= D_2 \\ y_{13} + y_{23} + y_{33} &= D_3 \\ y_{14} + y_{24} + y_{34} &= D_4 \\ y_{15} + y_{25} + y_{35} &= D_5 \end{aligned}\right\} \text{ demand} \\
& \left.\begin{aligned} x_{11} + x_{21} + x_{31} &= y_{11} + y_{12} + y_{13} + y_{14} + y_{15} \\ x_{12} + x_{22} + x_{32} &= y_{21} + y_{22} + y_{23} + y_{24} + y_{25} \\ x_{13} + x_{23} + x_{33} &= y_{31} + y_{32} + y_{33} + y_{34} + y_{35} \end{aligned}\right\} \text{ conservation} \\
& x_{11}, x_{12}, x_{13}, \\
& x_{21}, x_{22}, x_{23}, \\
& x_{31}, x_{32}, x_{33} \geq 0 \\
& y_{11}, y_{12}, y_{13}, y_{14}, y_{15}, \\
& y_{21}, y_{22}, y_{23}, y_{24}, y_{25}, \\
& y_{31}, y_{32}, y_{33}, y_{34}, y_{35} \geq 0
\end{aligned}$$

Code

See `problem_6.py`.

Solution

Objective	1963.82

Variable	Value	Reduced Cost	Objective Coef	Allowable Increase	Allowable Decrease
x[0,0]	93	0	1.52	0.03	∞
x[0,1]	0	0.03	1.6	∞	0.03
x[0,2]	0	0.03	1.4	∞	0.03
x[1,0]	0	0.12	1.7	∞	0.12
x[1,1]	86	0	1.63	0.12	0.06
x[1,2]	0	0.12	1.55	∞	0.12
x[2,0]	28	0	1.45	0.02	0.03
x[2,1]	0	0.07	1.57	∞	0.07
x[2,2]	67	0	1.3	0.03	0.02
y[0,0]	30	0	5.15	0.02	∞
y[0,1]	0	0.17	5.69	∞	0.17
y[0,2]	0	0.03	6.13	∞	0.03
y[0,3]	91	0	5.63	0.39	∞
y[0,4]	0	0.08	5.8	∞	0.08
y[1,0]	0	0.02	5.12	∞	0.02
y[1,1]	57	0	5.47	0.17	∞
y[1,2]	29	0	6.05	0.02	0.03
y[1,3]	0	0.54	6.12	∞	0.54
y[1,4]	0	0.04	5.71	∞	0.04
y[2,0]	0	0.02	5.32	∞	0.02
y[2,1]	0	0.49	6.16	∞	0.49
y[2,2]	19	0	6.25	0.03	0.02
y[2,3]	0	0.39	6.17	∞	0.39
y[2,4]	48	0	5.87	0.04	∞

Table 15: Variables

Name	Value	Shadow Price	RHS	Slack	Allowable Increase	Allowable Decrease
supply[0]	93	−0.06	93	0	28	2
supply[1]	86	0	88	2	∞	2
supply[2]	95	−0.13	95	0	29	2
demand[0]	30	6.73	30	0	2	28
demand[1]	57	7.1	57	0	2	57
demand[2]	48	7.68	48	0	2	29
demand[3]	91	7.21	91	0	2	28
demand[4]	48	7.3	48	0	2	29
balance[0]	0	1.58	0	0	2	28
balance[1]	0	1.63	0	0	2	86
balance[2]	0	1.43	0	0	2	29

Table 16: Constraints

Edge	Cost
x[0,0]	1.52
x[0,1]	1.6
x[0,2]	1.4
x[1,0]	1.7
x[1,1]	1.63
x[1,2]	1.55
x[2,0]	1.45
x[2,1]	1.57
x[2,2]	1.3
y[0,0]	5.15
y[0,1]	5.69
y[0,2]	6.13
y[0,3]	5.63
y[0,4]	5.8
y[1,0]	5.12
y[1,1]	5.47
y[1,2]	6.05
y[1,3]	6.12
y[1,4]	5.71
y[2,0]	5.32
y[2,1]	6.16
y[2,2]	6.25
y[2,3]	6.17
y[2,4]	5.87

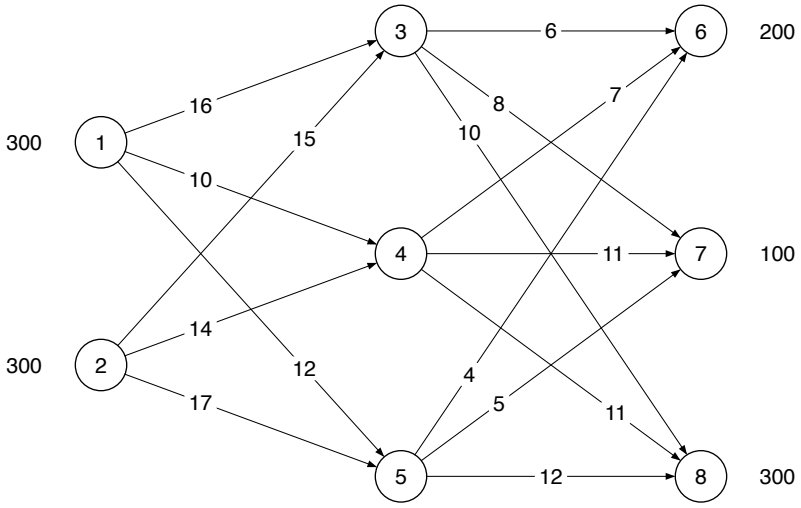
Table 17: Cost

Add the capacity to Well 3. Each extra TBD there replaces more expensive oil from Well 2 with cheaper oil from Well 3.

How much it's worth investing:

- Up to $0.13 per TBD of new capacity, in the same cost units as the tariffs.
- That rate holds for up to 29 TBD of extra capacity (the allowable increase), so Well 3 can go up to 124 TBD.
- At most, that adds up to 29 x $0.13 = $3.77 in total savings. That is the most we should pay, in the same per-day units as the tariffs.

Problem 7



Sets

- $I = \{1, 2\}$
- $J = \{3, 4, 5\}$
- $K = \{6, 7, 8\}$

Indices

- $i \in I$
- $j \in J$
- $k \in K$

Parameters

- $c_{ijk} = c_{ij} + c_{jk}$: unit cost of shipping from supplier i through transshipment node j to demand node k

$$C^{IJ} = \begin{bmatrix} 16 & 10 & 12 \\ 15 & 14 & 17 \end{bmatrix} \quad C^{JK} = \begin{bmatrix} 6 & 8 & 10 \\ 7 & 11 & 11 \\ 4 & 5 & 12 \end{bmatrix}$$

- S_i: supply of source node i

$$S = \begin{bmatrix} 300 \\ 300 \end{bmatrix}$$

- D_k: demand of sink node k

$$D = \begin{bmatrix} 200 \\ 100 \\ 300 \end{bmatrix}$$

Decision Variables

- x_{ijk}: quantity from supplier i to k going through j

Model

Old Model (x_{ij}, y_{jk})

$$
\begin{aligned}
\min \quad & \sum_{i\in I}\sum_{j\in J} c_{ij}x_{ij} + \sum_{j\in J}\sum_{k\in K} c_{jk}y_{jk} \\
\text{s.t.} \quad & \sum_{j\in J} x_{ij} \le S_i && \forall i \in I && \text{supply} \\
& \sum_{j\in J} y_{jk} \ge D_k && \forall k \in K && \text{demand} \\
& \sum_{i\in I} x_{ij} = \sum_{k\in K} y_{jk} && \forall j \in J && \text{conservation} \\
& x_{ij} \ge 0 && \forall i \in I, j \in J && \text{non-negativity} \\
& y_{jk} \ge 0 && \forall j \in J, k \in K && \text{non-negativity}
\end{aligned}
$$

$$
\begin{aligned}
\min \quad & c_{13}x_{13} + c_{14}x_{14} + c_{15}x_{15} + \\
& c_{23}x_{23} + c_{24}x_{24} + c_{25}x_{25} + \\
& c_{36}y_{36} + c_{37}y_{37} + c_{38}y_{38} + \\
& c_{46}y_{46} + c_{47}y_{47} + c_{48}y_{48} + \\
& c_{56}y_{56} + c_{57}y_{57} + c_{58}y_{58} \\
\text{s.t.} \quad & \left.\begin{aligned} x_{13} + x_{14} + x_{15} &\le S_1 \\ x_{23} + x_{24} + x_{25} &\le S_2 \end{aligned}\right\} \ \text{supply} \\
& \left.\begin{aligned} y_{36} + y_{46} + y_{56} &\ge D_6 \\ y_{37} + y_{47} + y_{57} &\ge D_7 \\ y_{38} + y_{48} + y_{58} &\ge D_8 \end{aligned}\right\} \ \text{demand} \\
& \left.\begin{aligned} x_{13} + x_{23} &= y_{36} + y_{37} + y_{38} \\ x_{14} + x_{24} &= y_{46} + y_{47} + y_{48} \\ x_{15} + x_{25} &= y_{56} + y_{57} + y_{58} \end{aligned}\right\} \ \text{conservation} \\
& x_{13}, x_{14}, x_{15}, x_{23}, x_{24}, x_{25} \ge 0 \\
& y_{36}, y_{37}, y_{38}, y_{46}, y_{47}, y_{48}, y_{56}, y_{57}, y_{58} \ge 0
\end{aligned}
$$

New Model (x_{ijk})

$$
\begin{aligned}
\min \quad & \sum_{i\in I}\sum_{j\in J}\sum_{k\in K} (c_{ij} + c_{jk})x_{ijk} \\
\text{s.t.} \quad & \sum_{j\in J}\sum_{k\in K} x_{ijk} \le S_i && \forall i \in I && \text{supply} \\
& \sum_{i\in I}\sum_{j\in J} x_{ijk} \ge D_k && \forall k \in K && \text{demand} \\
& x_{ijk} \ge 0 && \forall i \in I, j \in J, k \in K
\end{aligned}
$$

$$
\begin{aligned}
\min \quad & (c_{13} + c_{36})x_{136} + (c_{13} + c_{37})x_{137} + (c_{13} + c_{38})x_{138} + \\
& (c_{14} + c_{46})x_{146} + (c_{14} + c_{47})x_{147} + (c_{14} + c_{48})x_{148} + \\
& (c_{15} + c_{56})x_{156} + (c_{15} + c_{57})x_{157} + (c_{15} + c_{58})x_{158} + \\
& (c_{23} + c_{36})x_{236} + (c_{23} + c_{37})x_{237} + (c_{23} + c_{38})x_{238} + \\
& (c_{24} + c_{46})x_{246} + (c_{24} + c_{47})x_{247} + (c_{24} + c_{48})x_{248} + \\
& (c_{25} + c_{56})x_{256} + (c_{25} + c_{57})x_{257} + (c_{25} + c_{58})x_{258} \\
\text{s.t.} \quad & \left.\begin{aligned} x_{136} + x_{137} + x_{138} + x_{146} + x_{147} + \\ x_{148} + x_{156} + x_{157} + x_{158} \le S_1 \\ x_{236} + x_{237} + x_{238} + x_{246} + x_{247} + \\ x_{248} + x_{256} + x_{257} + x_{258} \le S_2 \end{aligned}\right\} \ \text{supply} \\
& \left.\begin{aligned} x_{136} + x_{146} + x_{156} + x_{236} + x_{246} + x_{256} &\ge D_6 \\ x_{137} + x_{147} + x_{157} + x_{237} + x_{247} + x_{257} &\ge D_7 \\ x_{138} + x_{148} + x_{158} + x_{238} + x_{248} + x_{258} &\ge D_8 \end{aligned}\right\} \ \text{demand} \\
& x_{136}, x_{137}, x_{138}, x_{146}, x_{147}, x_{148}, x_{156}, x_{157}, x_{158}, \\
& x_{236}, x_{237}, x_{238}, x_{246}, x_{247}, x_{248}, x_{256}, x_{257}, x_{258} \ge 0
\end{aligned}
$$

Code

See `problem_7.py`.

Solution

Objective	12400

Variable	Value	Reduced Cost	Objective Coef	Allowable Increase	Allowable Decrease
x[0,0,0]	0	6	22	∞	6
x[0,0,1]	0	7	24	∞	7
x[0,0,2]	0	6	26	∞	6
x[0,1,0]	0	1	17	∞	1
x[0,1,1]	0	4	21	∞	4
x[0,1,2]	0	1	21	∞	1
x[0,2,0]	200	0	16	1	0
x[0,2,1]	100	0	17	0	∞
x[0,2,2]	0	4	24	∞	4
x[1,0,0]	0	0	21	∞	0
x[1,0,1]	0	1	23	∞	1
x[1,0,2]	0	0	25	∞	0
x[1,1,0]	0	0	21	∞	0
x[1,1,1]	0	3	25	∞	3
x[1,1,2]	300	0	25	0	∞
x[1,2,0]	0	0	21	0	1
x[1,2,1]	0	0	22	∞	0
x[1,2,2]	0	4	29	∞	4

Table 18: Variables

Name	Value	Shadow Price	RHS	Slack	Allowable Increase	Allowable Decrease
supply[0]	300	−5	300	0	0	0
supply[1]	300	0	300	0	∞	0
demand[0]	200	21	200	0	0	0
demand[1]	100	22	100	0	0	0
demand[2]	300	25	300	0	0	300

Table 19: Constraints

Path	$k=0$	$k=1$	$k=2$
x[0,0,·]	22	24	26
x[0,1,·]	17	21	21
x[0,2,·]	16	17	24
x[1,0,·]	21	23	25
x[1,1,·]	21	25	25
x[1,2,·]	21	22	29

Table 20: Path costs c_{ijk}

Appendix

Problem 1

```python
import numpy as np
import gurobipy as gp
from gurobipy import GRB

S = np.array([6, 5, 6])
D = np.array([4, 8, 7])
M = np.array([
```

```
    [120, 180, 0],
    [300, 100, 80],
    [200, 250, 120],
])
C = 10
P = 5000

m = gp.Model("refineries")

x = m.addMVar(M.shape, lb=0, name="x")
s = m.addMVar(D.shape, name="s")

m.setObjective((C * M * x).sum() + (P * s).sum(), GRB.MINIMIZE)

m.addConstr(x.sum(axis=1) <= S, name="capacity")
m.addConstr(x.sum(axis=0) + s == D, name="demand")
m.addConstr(s[0] == 0, name="no_shortage_1")
m.addConstr(x[0, 2] == 0, name="no_arc_1_3")

m.optimize()
```

	A	B	C	D	E	F
1						
2		Area 1	Area 2	Area 3	Supply	
3	Ref 1	1,2	1,8	1000	6	
4	Ref 2	3	1	0,8	5	
5	Ref 3	2	2,5	1,2	6	
6	Dummy (Shortage)	1000	5	5	2	
7	Demand	4	8	7	19	
8						
9						
10		Area 1	Area 2	Area 3	Supply	Shipped
11	Ref 1	4	2	0	6	6
12	Ref 2	0	4	1	5	5
13	Ref 3	0	0	6	6	6
14	Dummy (Shortage)	0	2	0	2	2
15	Demand	4	8	7		
16	Received	4	8	7		
17						
18	Objetivo	30,4				
19						

Problem 2

```
import numpy as np
import gurobipy as gp
from gurobipy import GRB

S = np.array([150, 150])
R = np.array([3, 1, 2])
T = np.array([
    [0.8, 0.6],
    [0.3, 0.4],
    [0.5, 0.6],
])

m = gp.Model("products")

x = m.addMVar(T.shape, name="x")
y = m.addVar(name="y")

m.setObjective(y, GRB.MAXIMIZE)

m.addConstr(x.sum(axis=1) >= R * y, name="requirements")
m.addConstr((T * x).sum(axis=0) <= S, name="capacity")

m.optimize()
```

	A	B	C	D	E
1					
2		Dept A	Dept B	Parts per product	
3	Part 1	0,8	0,6	3	
4	Part 2	0,3	0,4	1	
5	Part 3	0,5	0,6	2	
6	Available (h)	150	150		
7					
8					
9					
10		Dept A	Dept B	Produced	Needed
11	Part 1	33,78	250,00	283,78	283,78
12	Part 2	94,59	0,00	94,59	94,59
13	Part 3	189,19	0,00	189,19	189,19
14					
15	Hours used	150,00	150,00		
16					
17					
18	Objetivo	94,59	<- final products (z)		
19					

Problem 3

```python
import numpy as np
import gurobipy as gp
from gurobipy import GRB

M = np.array([1500, 1700, 1300, 2700])
P = np.array([88, 15, 105, 95])
K = 12000
E0 = 600

m = gp.Model("plane")

x = m.addMVar(M.shape, name="x")
s = m.addMVar(M.shape, name="s")
e = m.addMVar(M.shape, name="e")

m.setObjective((P * x).sum(axis=0), GRB.MINIMIZE)

m.addConstr(s[0] == e[-1] + x[0], name="balance_0")
m.addConstr(s[1:] == e[:-1] + x[1:], name="balance")
m.addConstr(s - e == M * (1 + (s + e) / 4000), name="consumption")
m.addConstr(s <= K, name="capacity")
m.addConstr(e >= E0, name="reserve")

m.optimize()
```

	A	B	C	D	E	F	G	H
1								
2	Leg	From	To	Miles	Price ($/gal)			
3	1	LA	Houston	1500	0,88			
4	2	Houston	NYC	1700	0,15			
5	3	NYC	Miami	1300	1,05			
6	4	Miami	LA	2700	0,95			
7								
8	Tank capacity (gal)	12000						
9	Min end fuel (gal)	600						
10								
11								
12	Leg	Bought (B)	Start fuel (S)	End fuel (E)	Fuel used	S - Fuel used	Prev E + B	
13	1	3.120,00	3.720,00	600,00	3.120,00	600,00	3.720,00	
14	2	11.400,00	12.000,00	3.649,12	8.350,88	3.649,12	12.000,00	
15	3	0,00	3.649,12	877,86	2.771,27	877,86	3.649,12	
16	4	10.522,14	11.400,00	600,00	10.800,00	600,00	11.400,00	
17								
18								
19	Objetivo	$14.451,64	<- total fuel cost ($)					
20								

Problem 4

```python
import numpy as np
import gurobipy as gp
from gurobipy import GRB

K = np.array([90, 50, 80, 60])
D = np.array([120, 100, 110])
C = np.array([
    [70, 100, 50],
    [120, 90, 40],
    [70, 30, 110],
    [90, 50, 70],
])

m = gp.Model("steel")

m = gp.Model("steel")

x = m.addMVar(C.shape, name="x")

m.setObjective((C * x).sum(), GRB.MINIMIZE)

m.addConstr(x.sum(axis=1) == K, name="supply")
m.addConstr(x.sum(axis=0) <= D, name="demand")

m.optimize()
```

	A	B	C	D	E	F
1						
2		Richmond	Winston-Salem	Durham	Capacity	
3	Charlotte	70	100	50	90	
4	Raleigh	120	90	40	50	
5	Lexington	70	30	110	80	
6	Danville	90	50	70	60	
7	Demand	120	100	110	330	
8						
9						
10		Richmond	Winston-Salem	Durham	Capacity	Shipped
11	Charlotte	30	0	60	90	90
12	Raleigh	0	0	50	50	50
13	Lexington	40	40	0	80	80
14	Danville	0	60	0	60	60
15	Demand	120	100	110		
16	Received	70	100	110		
17	Unmet demand	50	0	0		
18						
19	Objetivo	$14.100	<- total shipping cost ($)			
20						

Problem 5

```python
import numpy as np
import gurobipy as gp
from gurobipy import GRB

H = np.array([
    [400, 500],
    [440, 550],
    [484, 605],
])
C = np.array([
    [8000, 10000],
    [8800, 11000],
    [9680, 12100],
])
D = np.array([
    [1200, 380],
    [2500, 1500],
    [1800, 1200]
])
K = np.array([
    4000,
    3500,
    2000,
])
E = np.array([
    300,
    200,
])
Y0 = np.array([100, 80])

m = gp.Model("steel_path")

x = m.addMVar(C.shape, name="x")
y = m.addMVar(H.shape, name="y")

m.setObjective((C * x + H * y).sum(), GRB.MINIMIZE)

m.addConstr(y[0] == Y0 + x[0] - D[0], name="balance_0")
m.addConstr(y[1:] == y[:-1] + x[1:] - D[1:], name="balance")
m.addConstr(x.sum(axis=1) <= K, name="capacity")
m.addConstr(y[-1] >= E, name="end_inventory")

m.optimize()
```

	A	B	C	D	E	F
1						
2		Quarter 1	Quarter 2	Quarter 3	Initial inv	Min final inv
3	Demand A21	1.200	2.500	1.800	100	300
4	Demand B32	380	1.500	1.200	80	200
5	Capacity	4.000	3.500	2.000		
6	Prod cost A21	8.000	8.800	9.680		
7	Prod cost B32	10.000	11.000	12.100		
8	Holding cost A21	400	440	484		
9	Holding cost B32	500	550	605		
10						
11						
12		Quarter 1	Quarter 2	Quarter 3		
13	Production A21	1.100	3.200	1.400		
14	Production B32	2.900	300	0		
15	Total production	4.000	3.500	1.400		
16	Capacity	4.000	3.500	2.000		
17						
18	End inv A21	0	700	300		
19	End inv B32	2.600	1.400	200		
20						
21	Objetivo	$85.456.200	<- total cost ($)			
22						

Problem 6

```python
import numpy as np
import gurobipy as gp
from gurobipy import GRB

S = np.array([93, 88, 95])
D = np.array([30, 57, 48, 91, 48])
C_ij = np.array([
    [1.52, 1.60, 1.40],
    [1.70, 1.63, 1.55],
    [1.45, 1.57, 1.30],
])
C_jk = np.array([
    [5.15, 5.69, 6.13, 5.63, 5.80],
    [5.12, 5.47, 6.05, 6.12, 5.71],
    [5.32, 6.16, 6.25, 6.17, 5.87],
])

m = gp.Model("steel")

x = m.addMVar(C_ij.shape, name="x")
y = m.addMVar(C_jk.shape, name="y")

m.setObjective((C_ij * x).sum() + (C_jk * y).sum(), GRB.MINIMIZE)

m.addConstr(x.sum(axis=1) <= S, name="supply")
m.addConstr(y.sum(axis=0) == D, name="demand")
m.addConstr(x.sum(axis=0) == y.sum(axis=1), name="balance")

m.optimize()
```

	A	B	C	D	E	F	G	H
1								
2		Pump A	Pump B	Pump C	Capacity			
3	Well 1	1,52	1,60	1,40	93			
4	Well 2	1,70	1,63	1,55	88			
5	Well 3	1,45	1,57	1,30	95			
6								
7		R1	R2	R3	R4	R5		
8	Pump A	5,15	5,69	6,13	5,63	5,80		
9	Pump B	5,12	5,47	6,05	6,12	5,71		
10	Pump C	5,32	6,16	6,25	6,17	5,87		
11	Demand	30	57	48	91	48		
12								
13								
14		Pump A	Pump B	Pump C	Shipped	Capacity		
15	Well 1	93	0	0	93	93		
16	Well 2	0	86	0	86	88		
17	Well 3	28	0	67	95	95		
18	Into pump	121	86	67				
19								
20								
21		R1	R2	R3	R4	R5	Out of pump	Into pump
22	Pump A	30	0	0	91	0	121	121
23	Pump B	0	57	29	0	0	86	86
24	Pump C	0	0	19	0	48	67	67
25	Received	30	57	48	91	48		
26	Demand	30	57	48	91	48		
27								
28	Objetivo	$1.963,82	<- total cost ($)					
29								

Problem 7

```python
import numpy as np
import gurobipy as gp
from gurobipy import GRB

S = np.array([300, 300])
D = np.array([200, 100, 300])
C_ij = np.array([
    [16, 10, 12],
    [15, 14, 17],
])
C_jk = np.array([
    [6, 8, 10],
    [7, 11, 11],
    [4, 5, 12],
])

C = C_ij[:, :, None] + C_jk[None, :, :]

m = gp.Model("steel_paths")

m.Params.OutputFlag = 0

x = m.addMVar(C.shape, name="x")

m.setObjective((C * x).sum(), GRB.MINIMIZE)

supply = m.addConstr(x.sum(axis=2).sum(axis=1) <= S, name="supply")
demand = m.addConstr(x.sum(axis=0).sum(axis=0) == D, name="demand")

m.optimize()
```

	A	B	C	D	E	F
3	Nebraska	16	10	12	300	
4	Colorado	15	14	17	300	
5						
6	Outbound cost	Chicago	St. Louis	Cincinnati		
7	Kansas City	6	8	10		
8	Omaha	7	11	11		
9	Des Moines	4	5	12		
10	Demand	200	100	300		
11						
12	Path cost (i-j)	Chicago	St. Louis	Cincinnati		
13	Nebraska - Kansas City	22	24	26		
14	Nebraska - Omaha	17	21	21		
15	Nebraska - Des Moines	16	17	24		
16	Colorado - Kansas City	21	23	25		
17	Colorado - Omaha	21	25	25		
18	Colorado - Des Moines	21	22	29		
19						
20						
21	Flow X(i,j,k)	Chicago	St. Louis	Cincinnati	Total	
22	Nebraska - Kansas City	0	0	0	0	
23	Nebraska - Omaha	0	0	0	0	
24	Nebraska - Des Moines	200	100	0	300	
25	Colorado - Kansas City	0	0	300	300	
26	Colorado - Omaha	0	0	0	0	
27	Colorado - Des Moines	0	0	0	0	
28	Received	200	100	300		
29	Demand	200	100	300		
30						
31		Shipped	Supply			
32	Nebraska	300	300			
33	Colorado	300	300			
34						
35	Objetivo	$12.400	<- total cost ($)			
36						